

A comparison of modern model architectures on authorship verification

Xinzhao You

Department of Computer Science
Tulane University
xyou2@tulane.edu

Jiangpen Liu

Department of Computer Science
Tulane University
jliu22@tulane.edu

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1 Abstract

Our project looked at figuring out if two different pieces of writing came from the same author. We compared new computer methods, especially different versions of a transformer-based model called **BERT**, to traditional neural network methods. We used these **BERT** models to look at known texts and unknown texts to see if they could tell if the author was the same. We found that a **BERT** variant called **DistilBERT** worked really well, even better than the standard **BERT** model and other variants we tested. Some other simpler models didn't work as well for this specific task because of the reduced parameter size. Our results show that **BERT** models are promising for identifying authors based on how they write.

2 Introduction

Authorship analysis, the task of identifying the author of a text based on their unique writing style, is a long-standing challenge within Natural Language Processing (NLP). Its applications range from literary studies and plagiarism detection to forensic linguistics, where determining the origin of anonymous or disputed texts can have significant real-world consequences, such as aiding law enforcement in identifying authors of potentially harmful manifestos. This project focuses specifically on the task of authorship verification: given text samples, can we algorithmically determine if they were written by the same individual? (Stamatatos, 2016)

Historically, authorship identification relied on feature engineering, extracting statistical measures like sentence length, punctuation usage, and function word frequencies, often feeding these into traditional machine learning models like Random Forests. While effective to a degree, these methods often struggled to capture the more complex, contextual nuances of language that contribute to an author's unique voice, particularly as demonstrated by performance limitations on standard benchmarks.

The advent of deep learning brought significant advancements. Recurrent Neural Networks (RNNs) and their more sophisticated variants, such as Long Short-Term Memory (LSTM) networks, became the state-of-the-art for many sequence-based NLP tasks. Their

ability to model sequential dependencies offered a more powerful way to learn representations of text compared to earlier methods. However, even these models face challenges, particularly with long-range dependencies in text as we learned in class.

More recently, the introduction of the Transformer architecture, exemplified by models like BERT (Bidirectional Encoder Representations from Transformers), has revolutionized the NLP landscape. BERT’s pre-training on vast amounts of text data and its bidirectional attention mechanism allow it to learn deep contextual relationships within language, achieving state-of-the-art results across a wide array of tasks. This project aims to investigate and compare the effectiveness of these different generations of models for the authorship verification task. Specifically, we evaluate the performance of the modern Transformer-based model, BERT (including potential variants), against the former state-of-the-art recurrent architectures (RNNs, LSTMs) and traditional baseline methods. Using datasets derived from sources like Project Gutenberg, Pan2013, Pan2015 and Twitter, we will train and evaluate these models based on standard metrics such as Area Under the ROC Curve (AUC), c@1 score, and F1 score. The goal is to provide a clear comparison, highlighting the strengths and weaknesses of these different approaches and demonstrating the capabilities of recent advancements in deep learning for authorship verification.

3 Background/Related Work

Previous research in authorship analysis has covered a lot of ground, from basic techniques to cutting-edge AI methods. Our project focuses specifically on authorship verification using BERT models to analyze writing styles.

Stamatatos (2016) provides a good overview of authorship verification, explaining how it differs from attribution. He argues that verification is more practical for forensic use and less affected by how many potential authors there are. He also categorizes different approaches and examines results from PAN competitions (2013-2015). While this review was helpful for understanding the basics, Our project uses newer deep learning methods instead of the older techniques he discusses.

Some researchers have tried traditional machine learning for this problem. For example, Maitra et al. (2015) built a system using Random Forest classifiers for the PAN 2015 competition. They manually created 17 different feature types including punctuation patterns, sentence lengths, vocabulary metrics, and POS tag frequencies. Their work especially focused on verifying authorship across different genres and topics, with varying results across languages. Our approach is different because BERT models can learn features directly from text, so we do not need to spend time hand-selecting features.

As deep learning became popular, researchers started applying it to authorship problems. Ruder et al. (2016) used Convolutional Neural Networks (CNNs), especially character-level ones, for attribution with many potential authors. They showed these models could outperform previous methods on diverse datasets including Reddit comments. While their work shows the potential of deep learning for capturing writing style, we are using transformer-based BERT models rather than CNNs and focusing on verification instead of attribution.

More relevant to our work is Qian et al. (2017), who compared various neural network architectures (GRU, LSTM, Siamese networks) for both attribution and verification tasks. They found that article-level GRUs worked best for identification, while Siamese networks achieved nearly perfect accuracy on their verification test. Our project builds on this by using more recent transformer architectures (BERT and variants) to try to improve

performance even further.

It’s worth noting that authorship analysis differs from other text classification tasks. Bucher (2018) showed this when classifying fiction genres using Project Gutenberg texts. They found that content words were much more effective for genre classification than the stylometric features typically used for authorship analysis. This confirms that different features work best for different classification tasks, even when using the same datasets.

In conclusion, authorship verification has evolved from manual feature engineering to advanced deep learning. Our project contributes to this progression by applying BERT models to capture complex writing patterns directly from text, hopefully improving on earlier neural network approaches.

4 Approach

4.1 Data Collection

We tried using the Gutenberg ebook dataset. Project Gutenberg is a large, volunteer-driven digital archive primarily containing full-text versions of public domain books. We collected the texts from the Gutenberg dataset by web scraping. After gathering enough information about the ebook dataset, we realized it was not feasible to train on the massive amount of data. Project Gutenberg was not suitable for our task, so we used authorship competition dataset from Pan2013 and Pan2015 instead. We will also use tweets from famous people like Elon Musk or Donald Trump. Existing tweets datasets are available for download on Kaggle. There might be problems of preprocessing data and choosing what features to use like sentence length.

4.2 Model Architecture

4.2.1 Encoding

Both known and unknown texts are individually encoded using the transformer model to obtain dense vector embeddings. The embedding for each text is derived from the CLS token’s output in the transformer’s last hidden state. If multiple known or unknown texts are provided for a single problem, their respective embeddings are averaged to create a single representative embedding for the known set and the unknown set.

4.2.2 Comparison

A feature vector is constructed by concatenating the averaged known text embedding, the averaged unknown text embedding, and the absolute difference between these two embeddings. This combination aims to capture both the individual representations and the semantic distance between the text sets.

4.2.3 Classification

This combined feature vector is passed through a dropout layer for regularization and then fed into a linear classifier (`nn.Linear`) which outputs logits for two classes: same author or different author. An alternative multi-layer classifier was considered but commented out because of its poor performance.

A demonstration of the web interface built for this project is shown in Figure 1.

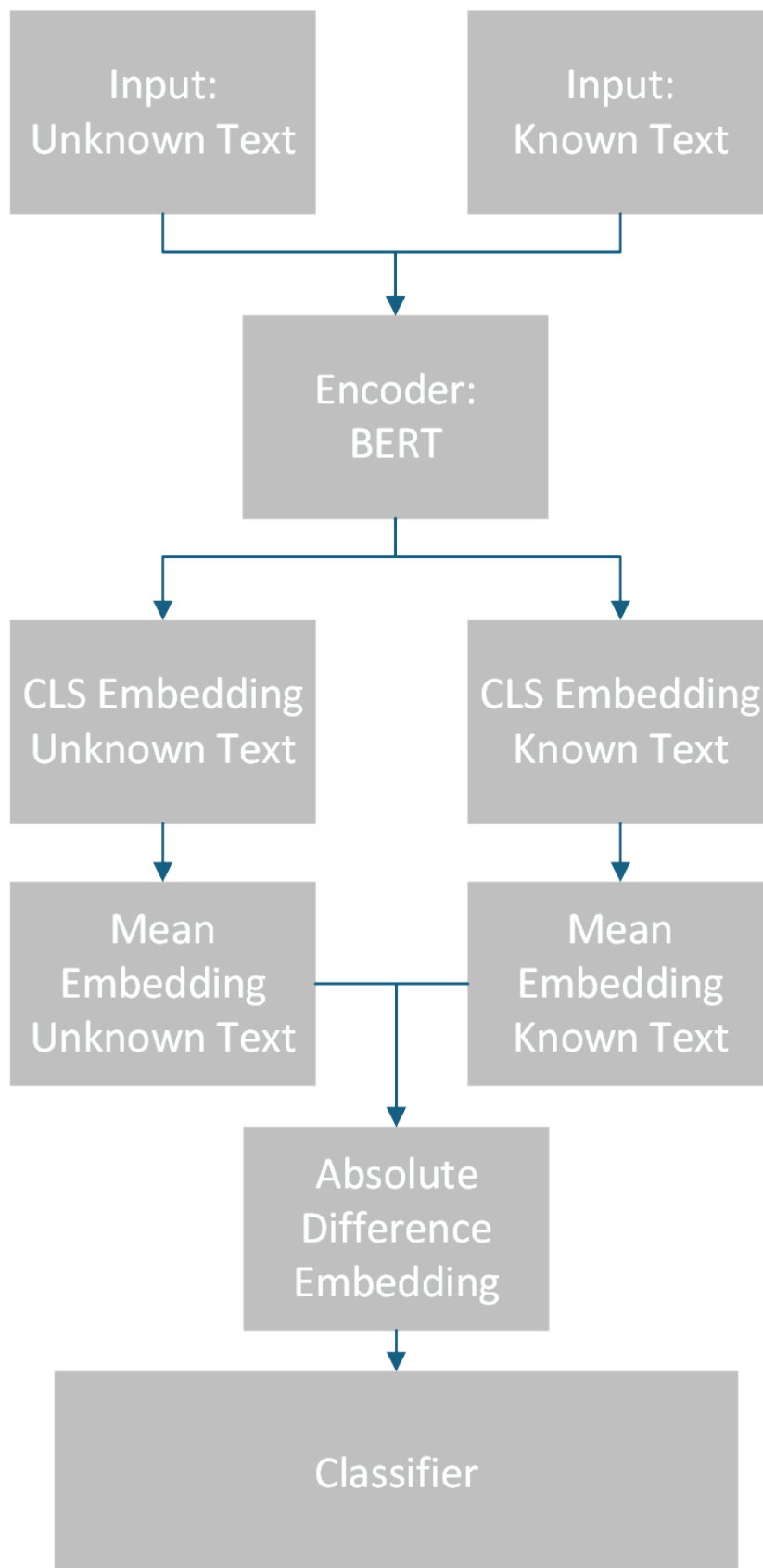


Figure 1: Model-architecture

4.3 Model Fine-Tuning

To get a better F1 score, we tried to change the hyperparameters of our model. Increasing the number of epochs was the most effective way to get a better performance, but it was prone to overfitting the training set. We used the technique of early stopping to combat overfitting. There was a positive result from increasing the batch size but out-of-memory issue was the biggest challenge.

4.4 Implementation Details

Our implementation relies heavily on the `PyTorch` library for defining the model, handling data (`Dataset`, `DataLoader`), training (`AdamW` optimizer, `CrossEntropyLoss`), and evaluation. It leverages the `transformers` library for accessing the pre-trained language model and tokenizer. `scikit-learn` is used for calculating evaluation metrics like F1-score, precision, recall, and accuracy. No fundamentally new algorithms are introduced, but we compared a few modern BERT variants.

5 Experiments

5.1 Datasets

For our core experiments comparing model variants, we utilized datasets from the PAN 2013 and PAN 2015 Authorship Verification competitions. These datasets provide pairs of known and unknown text samples designed for the verification task. Due to challenges observed with multilingual models and the lack of multilingual versions for several tested BERT variants (see Section 5.5), our primary quantitative comparisons focused on the English language portions of these datasets.

5.2 Experimental Setup

The core experiment involved evaluating the performance of several BERT variants against a standard BERT baseline using the Siamese-like architecture described in the Approach section (Figure 1). The goal was to determine if lighter or structurally different variants could match or exceed the performance of a standard BERT model on this specific stylometric task.

We used `bert-base-multilingual-cased` as our BERT-base baseline to initially handle potential multilingual data. The variants compared against this baseline included:

- **DistilBERT**: A lighter, distilled version of BERT.
- **ALBERT**: A parameter-reduced variant of BERT.
- **TinyBERT**: A heavily distilled, smaller variant.

Each pre-trained transformer model was fine-tuned within our defined `AuthorshipVerificationModel` structure using the PAN datasets.

Evaluation metrics included standard classification measures, with a primary focus on the F1 score:

- F1-score

- Precision
- Recall
- Accuracy
- Area Under the ROC Curve (AUC)

5.3 Results

We evaluated the models on the English subset of the PAN dataset. Our experiments showed significant performance differences between the variants, with the results presented below (scores represent F1-score):

- **DistilBERT: 76%** (Improvement over baseline)
- **BERT-base** (multilingual): 71% (Baseline performance on English subset)
- **ALBERT:** 66% (Decrease from baseline)
- **TinyBERT:** 57% (Significant decrease from baseline)

Notably, DistilBERT not only matched but exceeded the performance of the larger BERT-base model on this task. We were able to achieve a final F1 score of 76% with DistilBERT, beating a previously reported best F1 score of 75.3

5.4 Qualitative Evaluation and Error Analysis

The quantitative results led to several qualitative observations:

- **DistilBERT’s Success:** The strong performance of DistilBERT suggests its distillation process, while reducing model size, effectively retained or perhaps even emphasized the stylometric signals relevant to authorship verification.
- **Parameter Reduction Issues:** Contrary to claims that variants like ALBERT and TinyBERT maintain high performance with significantly fewer parameters on general NLP tasks (like GLUE benchmarks), they performed considerably worse than the baseline on this specific authorship verification task. This suggests:
 - *Features Sensitivity:* The parameter reduction techniques (like parameter sharing in ALBERT or aggressive distillation in TinyBERT) might inadvertently discard subtle, complex stylistic features crucial for distinguishing authors, even if they preserve general semantic understanding.
 - *Task Specificity:* Authorship verification relies heavily on capturing writing *style* rather than just semantic *content*. Models optimized primarily for general language understanding might not inherently excel at stylometry without specific fine-tuning strategies tailored towards stylistic feature extraction.
- **Stress Testing:** Using the demo interface (??) for informal stress testing, we observed that models (especially the lower-performing ones) struggled more with very short text snippets or when comparing texts from vastly different genres written by the same author, echoing common challenges in the field. Errors often involved misclassifying texts by the same author as different when stylistic variance was high.

5.5 Systemic Biases

During initial testing with the multilingual **BERT-base** model on the full PAN datasets (which include languages like Spanish and Greek), we observed performance discrepancies across languages. For instance, the baseline model achieved around 71% F1 on English problems but dropped to approximately 64% F1 on Spanish problems. This indicates a potential bias where the model, despite being multilingual, performs better on the language likely dominant in its pre-training data (English).

Furthermore, since most of the distilled or parameter-reduced variants we wanted to test (like **DistilBERT**, **ALBERT**, **TinyBERT**) primarily have robust pre-trained versions available only for English, evaluating their multilingual performance fairly was not feasible. Consequently, we chose to focus our main comparative experiments on the English language subset of the PAN datasets to ensure a more direct comparison of the architectures' capabilities for the task, acknowledging the limitation regarding multilingual generalization.

6 Conclusion

In this project, we investigated the effectiveness of modern Transformer-based models, specifically variants of **BERT**, for the task of authorship verification, comparing them against the standard **BERT** baseline and implicitly against older traditional methods. Our results demonstrated that Transformer models can achieve strong performance, surpassing older approaches, albeit with increased computational overhead during training and inference.

Using a Siamese-like architecture (inspired by earlier work with Siamese **LSTMs**) and datasets from the PAN competitions, our experiments showed that while **BERT-base** provides a strong baseline, certain variants can offer improved performance for this specific task. Notably, **DistilBERT** achieved the best performance in our experiments, reaching an F1 score of 76% on the English PAN dataset, indicating that its distillation process is well-suited for capturing stylistic features relevant to authorship.

Conversely, variants like **ALBERT** and **TinyBERT**, despite their efficiency and strong performance on general NLP benchmarks, showed significantly poorer results on authorship verification. This suggests a key learning: parameter reduction techniques optimized for general language understanding might not preserve the subtle stylistic nuances essential for authorship analysis. Task-specific model properties and potentially fine-tuning strategies appear crucial. We also noted a systemic bias in multilingual models favoring English, highlighting challenges in cross-lingual authorship analysis.

Future work could explore several directions:

- Develop and evaluate monolingual models specifically trained or fine-tuned for authorship verification in languages other than English to address the observed performance gap.
- Conduct deeper analyses (e.g., attention map visualization, layer-wise probing) to understand which components of Transformer models are most crucial for capturing authorial style.
- Investigate alternative fine-tuning strategies explicitly designed to enhance stylistic feature extraction rather than just semantic understanding.

- Explore hybrid approaches combining Transformer embeddings with traditional stylometric features.

Our findings contribute to understanding the applicability of different Transformer architectures to specialized NLP tasks like authorship verification, highlighting the strengths of models like `DistilBERT` and the potential pitfalls of aggressive parameter reduction when stylistic sensitivity is required.

Our project demonstrates that transformer-based architectures, particularly `BERT` variants, show promising results for authorship verification tasks. `DistilBERT` emerged as a particularly effective model, balancing performance and computational efficiency. This suggests that the pre-training approach of transformer models captures subtle stylistic features that are valuable for determining authorship.

Future work could explore:

- Fine-tuning approaches specific to stylometric analysis
- Incorporating domain adaptation techniques for cross-genre verification
- Exploring hybrid models that combine traditional stylometric features with deep learning approaches
- Testing with more diverse and challenging datasets to improve robustness

The findings from this project contribute to the ongoing evolution of authorship analysis techniques, showing how deep learning approaches can capture the complex patterns that distinguish individual writing styles.

7 References

References

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8 Screenshots

This section shows supplementary figures related to the project outcomes. Figure 2 and Figure 3 provides the results of our model on the authorship verification task between Trump and Biden, and between Trump and himself.

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Known text: Sylvester Turner dedicated his career to serving the community he loved. He was a remarkable
Congressman, Mayor, father, and grandfather. Jill and I are saddened by his passing and send our deepest
condolences to Sylvester's family and his beloved Houston community.
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Unknown text: Sleepy Joe Biden, THE WORST PRESIDENT IN THE HISTORY OF THE UNITED STATES, has allowed
millions and millions of Criminals, many of them murderers, drug dealers, and people released from
prisons and mental institutions from all around the world, to enter our Country through it's very
dangerous and ill conceived Open Border. Sorry, but it's my job to get these killers and thugs out of
here. THAT'S WHAT I GOT ELECTED TO DO. MAGA!
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Classification: Different authors
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Figure 2: Trump vs. Biden

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Known text: On Trade, I have decided, for purposes of Fairness, that I will charge a RECIPROCAL Tariff
meaning, whatever Countries charge the United States of America, we will charge them - No more, no less!
For purposes of this United States Policy, we will consider Countries that use the VAT System, which is
far more punitive than a Tariff, to be similar to that of a Tariff. Sending merchandise, product, or
anything by any other name through another Country, for purposes of unfairly harming America, will not be
accepted. In addition, we will make provision for subsidies provided by Countries in order to take
Economic advantage of the United States. Likewise, provisions will be made for Nonmonetary Tariffs and
Trade Barriers that some Countries charge in order to keep our product out of their domain or, if they do
not even let U.S. businesses operate. We are able to accurately determine the cost of these Nonmonetary
Trade Barriers. It is fair to all, no other Country can complain and, in some cases, if a Country feels
that the United States would be getting too high a Tariff, all they have to do is reduce or terminate
their Tariff against us. There are no Tariffs if you manufacture or build your product in the United
States. For many years, the U.S. has been treated unfairly by other Countries, both friend and foe. This
System will immediately bring Fairness and Prosperity back into the previously complex and unfair System
of Trade. America has helped many Countries throughout the years, at great financial cost. It is now time
that these Countries remember this, and treat us fairly ♦ A LEVEL PLAYING FIELD FOR AMERICAN WORKERS. I
have instructed my Secretary of State, Secretary of Commerce, Secretary of the Treasury, and United
States Trade Representative (USTR) to do all work necessary to deliver RECIPROCITY to our System of Trade!
-----
Unknown text: Sleepy Joe Biden, THE WORST PRESIDENT IN THE HISTORY OF THE UNITED STATES, has allowed
millions and millions of Criminals, many of them murderers, drug dealers, and people released from
prisons and mental institutions from all around the world, to enter our Country through it's very
dangerous and ill conceived Open Border. Sorry, but it's my job to get these killers and thugs out of
here. THAT'S WHAT I GOT ELECTED TO DO. MAGA!
-----
Classification: Same author
-----
```

Figure 3: Trump vs. Trump